

Perfect — let's turn that flat list into a **taxonomy tree of analog & mixed-signal circuits**. This way you see how a single MOSFET stage grows into full AMS subsystems.



Hierarchical Classification of Circuits

1. Basic Device-Level & Building Blocks

- **MOS/BJT Stages**
 - Common-Source Amplifier (resistive load, diode load, current-source load, degeneration)
 - Source Follower (Common Drain)
 - Common-Gate Amplifier
 - **Cascode Circuits**
 - Cascode Gain Stage
 - Folded Cascode Amplifier
 - Regulated Cascode
 - **Current Circuits**
 - Current Sources (basic, cascode, high-swing)
 - Current Mirrors (basic, cascode, high-swing, mismatch-aware)
 - **Switch Circuits**
 - Complementary MOS Switch
 - Dummy Switch (pedestal error reduction)
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2. Amplifiers

- **Op-Amps**

- o Single-Stage Op-Amps
 - o Two-Stage Op-Amps
 - o Telescopic Op-Amps
 - o Folded Cascode Op-Amps
 - o Miller-Compensated Op-Amps
 - **Differential Amplifiers**
 - o Single-Stage Differential Pair
 - o Two-Stage Differential Amplifier
 - o Differential-to-Single-Ended Converter
 - **Auxiliary / Special Amplifiers**
 - o Pre-Amplifiers (resistive load, NMOS load, PMOS load, current-source load, positive feedback)
 - o Sensor/Pixel Readout Amplifiers (column amplifiers for imagers, piezo interfaces)
 - o LNAs (Low Noise Amplifiers for RF)
 - o Power Amplifiers (RF PAs, efficiency classes)
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3. Data Converter Building Blocks

- **Sampling & Hold Circuits**
 - o Sample-and-Hold (SHA, flip-around, switched-capacitor)
 - o Aperture Jitter Models
- **Comparators & Latches**
 - o Static Comparator
 - o Dynamic Comparator
 - o Semi-Dynamic / Half-Latch Comparator

- o Pre-Amplifier Based Comparator
 - o Offset-Cancelled Comparator
 - **DAC Elements**
 - o Binary Weighted DACs (current-steering, voltage-mode)
 - o R-2R DACs (voltage-mode, current-mode)
 - o String DACs (resistor string)
 - o Segmented DACs (unary + binary weighted)
 - o Current Source DACs
 - o Capacitor DACs (C-DAC, charge redistribution, parasitic-insensitive)
 - **ADC Support Circuits**
 - o Charge Redistribution Networks
 - o Reference Voltage Buffers
 - o Output Offset Cancellation
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4. Converter Architectures

- **ADC Architectures**
 - o Flash ADC
 - o Successive Approximation Register (SAR) ADC
 - o Time-Interleaved SAR ADC
 - o Pipeline ADC
 - o Oversampled $\Delta\Sigma$ (Sigma-Delta) ADC
 - o Hybrid ADCs (e.g., pipeline-SAR)
- **DAC Architectures**
 - o R-2R DAC

- o String DAC
 - o Segmented DAC
 - o Current-Steering DAC
 - **Sigma-Delta ($\Delta\Sigma$) Modulators**
 - o First-Order
 - o Second-Order
 - o MASH (Multi-Stage Noise Shaping)
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5. Filters & Signal Processing

- **Active RC Filters**
 - **Switched-Capacitor Integrators / Filters**
 - **Digital Decimation Filters**
 - o Sinc
 - o FIR
 - o Hogenaur Structure
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6. Frequency Generation & Timing

- **Oscillators**
 - o Ring Oscillator
 - o LC Oscillator (cross-coupled, negative-gm)
 - o Crystal Oscillator
- **Voltage Controlled Oscillators (VCOs)**
- **Phase-Locked Loops (PLLs)**
 - o Type-I PLL

- o Type-II PLL
 - **Clock Generation & Distribution Circuits**
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7. References & Regulators

- **Voltage References**
 - o Supply-Independent References
 - o Bandgap References
 - **Current References**
 - **Voltage Regulators**
 - o Low Dropout Regulators (LDOs)
 - o NMOS Pass-Gate Regulators
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8. RF & Microwave Circuits

- **Mixers**
 - o Single-Balanced
 - o Double-Balanced (Gilbert Cell)
 - o Passive & Active Mixers
- **Frequency Converters**
 - o Downconverters / Upconverters
- **RF Front-Ends**
 - o Duplexers
 - o Matching Networks
 - o RF Filters
- **Passive RF Elements**

- o Transmission Lines (lumped & distributed models)
 - o Spiral Inductors (integrated with Q-factor considerations)
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9. Noise & Analysis Circuits

- **Device Noise Circuits**
 - o Thermal Noise
 - o Flicker Noise
 - o Shot Noise
 - **Capacitor Noise**
 - o kT/C Noise
 - **Feedback & Noise Analysis Models**
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✨ This hierarchy shows the **evolution path**:

MOS device → amplifier stage → op-amp → converter → PLL/RF front-end → full mixed-signal subsystem.

Would you like me to **turn this into a graphical “taxonomy map”** (like a tree diagram from primitive → complex circuits)? That way you could literally *see* how everything nests together.